Beyond Quantum Resilience
Embedded Security with & for PQC

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QUANTUM CLOUD —

How IBM's new five-qubit universal quan computer works

IBM achie

CHRIS LEE - 5/4

Intel Delivers 17-Qubit Superconducting Chip with Advanced Packaging to QuTech

NEWS | 23 October 2019

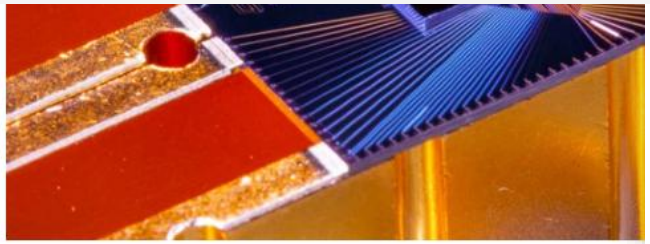
Hello quantum world! Goo

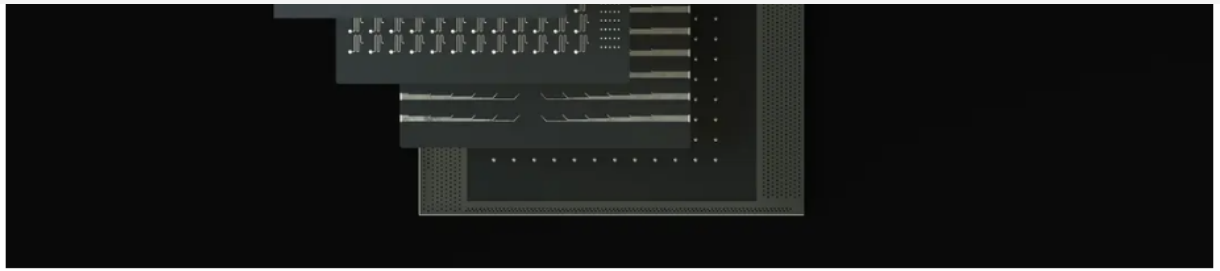
Eagle's quantum performance

Development Roadmap

IBM **Quantum**

	2016–2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2033+
	Run quantum circuits on the IBM Quantum Platform	Release multi-dimensional roadmap publicly with initial aim focused on scaling	Enhancing quantum execution speed by 100x with Qiskit Runtime	Bring dynamic circuits to unlock more computations	Enhancing quantum execution speed by 5x with quantum serverless and Execution modes	Improving quantum circuit quality and speed to allow 5K gates with parametric circuits	Enhancing quantum execution speed and parallelization with partitioning and quantum modularity	Improving quantum circuit quality to allow 7.5K gates	Improving quantum circuit quality to allow 10K gates	Improving quantum circuit quality to allow 15K gates	Improving quantum circuit quality to allow 100M gates	Beyond 2033, quantum-centric supercomputers will include 1000's of logical qubits unlocking the full power of quantum computing
Data Scientist						Platform						
						Code assistant	Functions	Mapping Collection	Specific Libraries			General purpose QC libraries
Researchers					Middleware							
					Quantum Serverless	Transpiler Service	Resource Management	Circuit Knitting x P	Intelligent Orchestration			Circuit libraries
Quantum Physicist			Qiskit Runtime									
	IBM Quantum Experience		QASM3	Dynamic circuits	Execution Modes	Heron (5K)	Flamingo (5K)	Flamingo (7.5K)	Flamingo (10K)	Flamingo (15K)	Starling (100M)	Blue Jay (1B)
						Error Mitigation	Error Mitigation	Error Mitigation	Error Mitigation	Error Mitigation	Error correction	Error correction
						5k gates 133 qubits	5k gates 156 qubits	7.5k gates 156 qubits	10k gates 156 qubits	15k gates 156 qubits	100M gates 200 qubits	1B gates 2000 qubits
						Classical modular	Quantum modular	Quantum modular	Quantum modular	Quantum modular	Error corrected modularity	Error corrected modularity
						133x3 = 399 qubits	156x7 = 1092 qubits	156x7 = 1092 qubits	156x7 = 1092 qubits	156x7 = 1092 qubits		
	Early: Canary 5 qubits; Albatross 16 qubits; Penguin 20 qubits; Prototype 53 qubits	Falcon: Benchmarking 27 qubits		Eagle: Benchmarking 127 qubits								

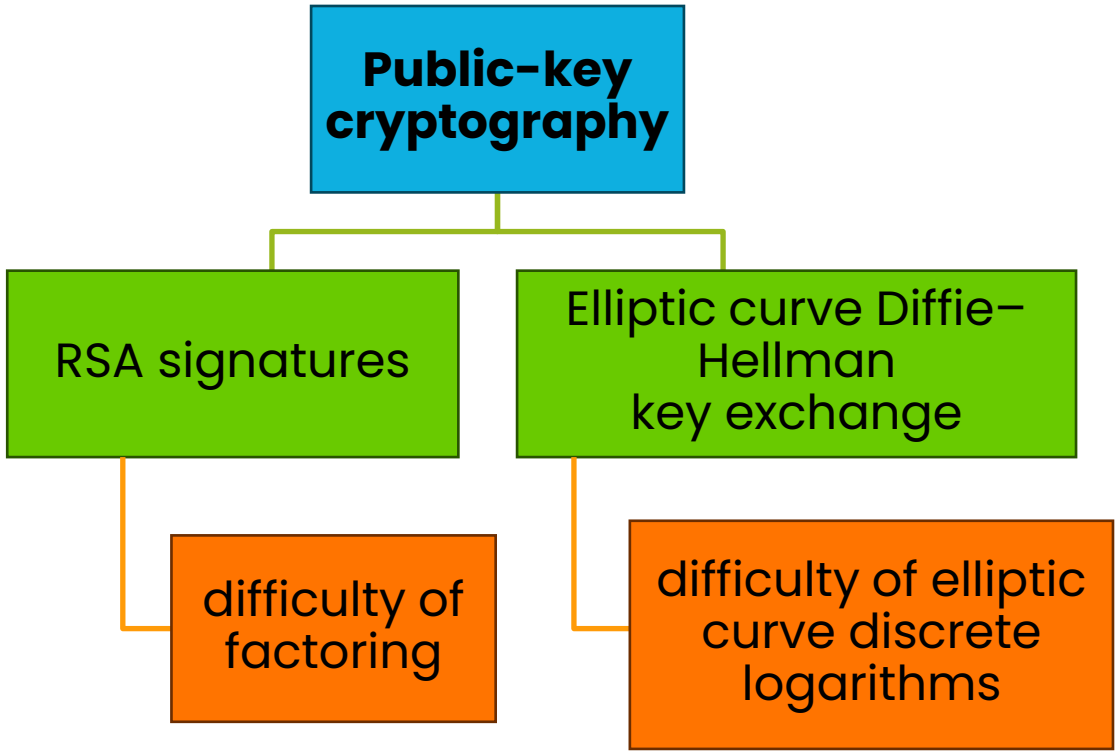




The Sycamore chip is composed of 54 qubits, each made of superconducting

IBM

Contemporary Public-Key Cryptography



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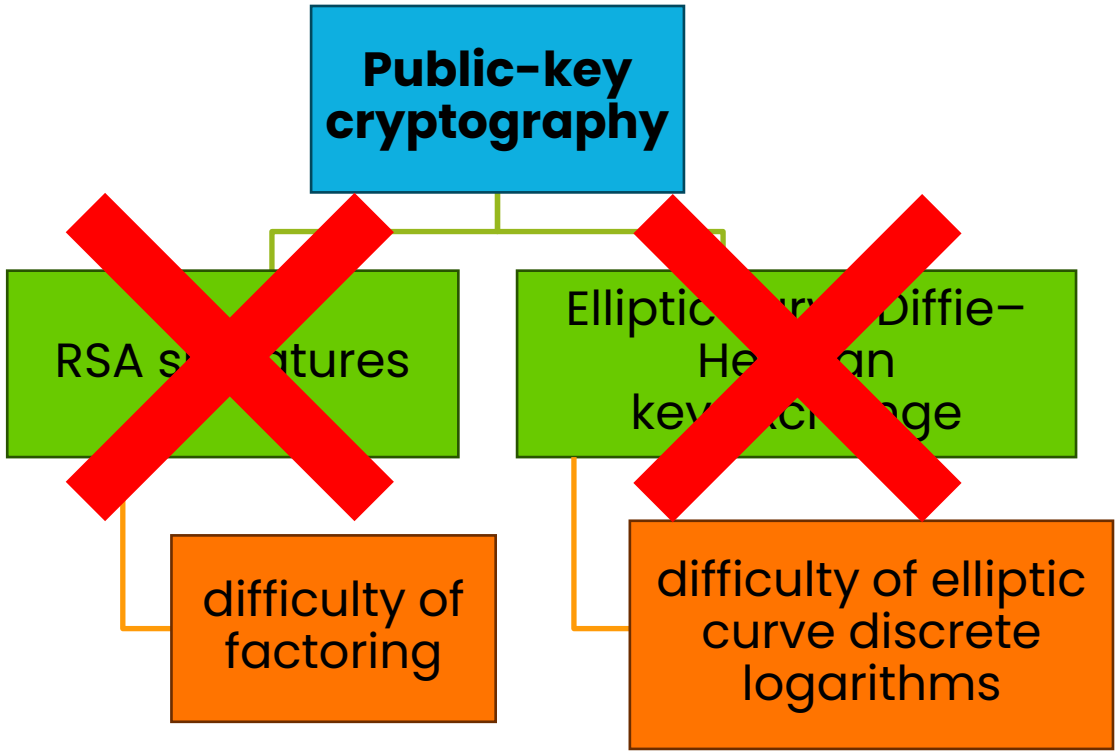


Shor's algorithm (1994)

What is the impact on the billions of embedded devices?

Automotive

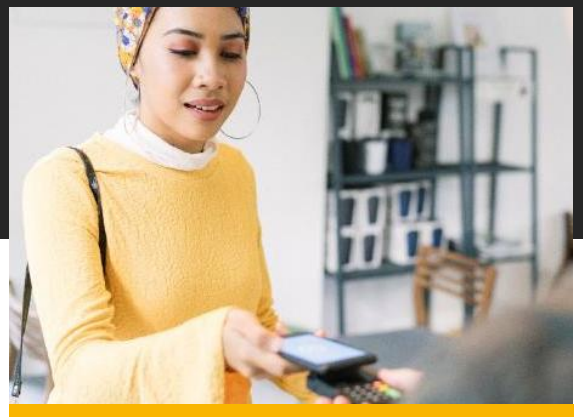
70%

70% connected cars by 2025

Industrial & IoT

12B

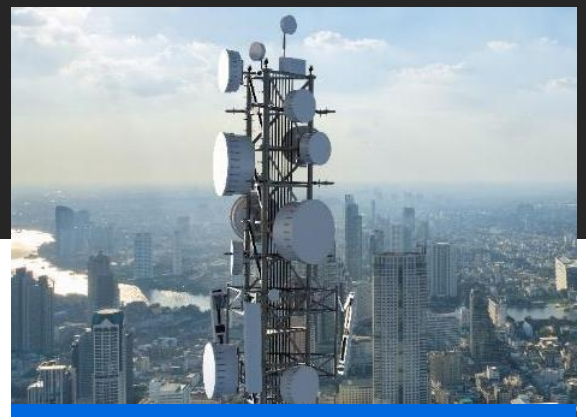
IoT Edge & end nodes from **6B units** in 2021 to **12B units** in 2025



Mobile

60B

Tagging **60B products** per year by 2025



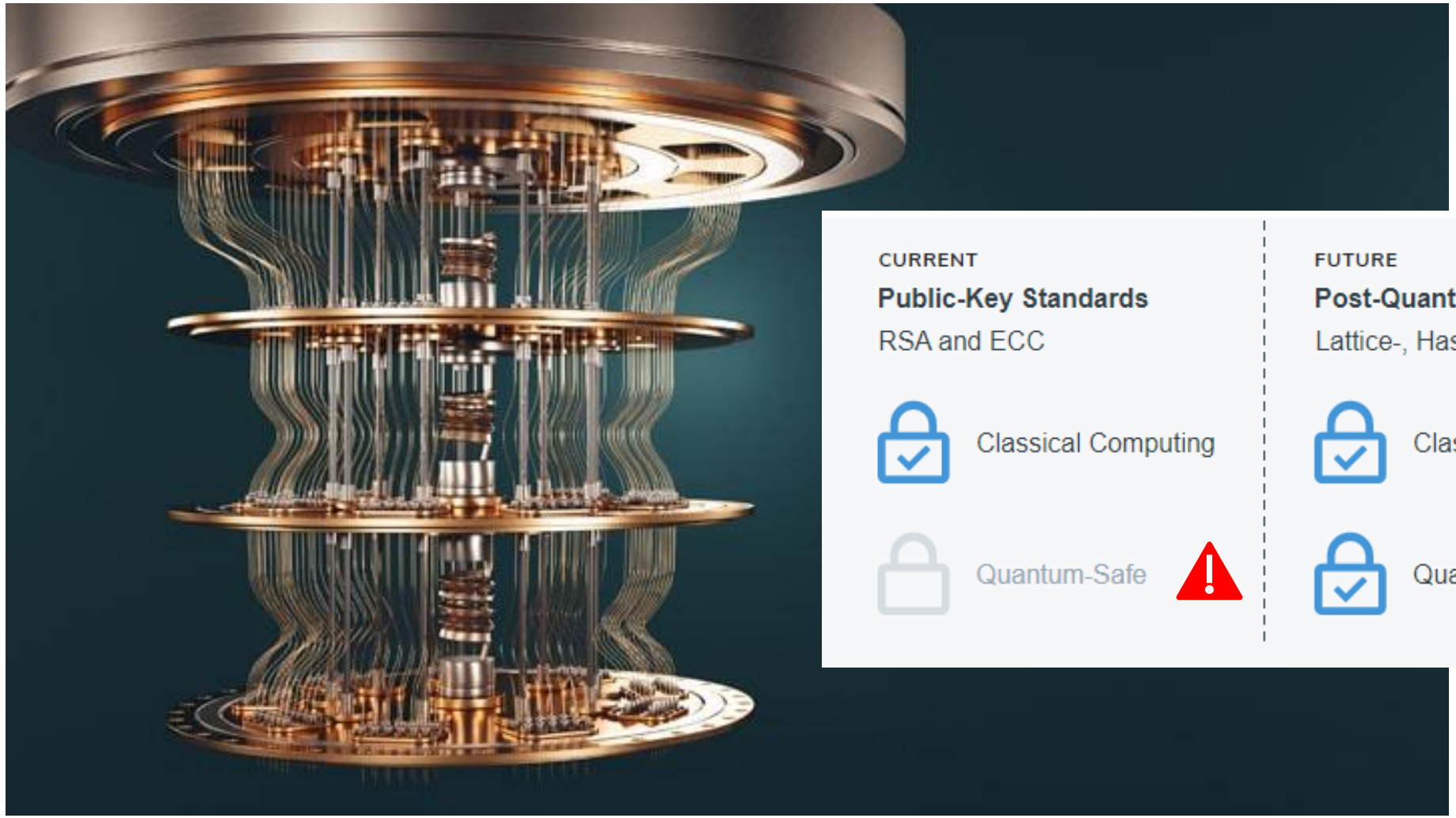
Communication Infrastructure

40B

Secure anchors & services for **40B processors**

Automotive | **eGovernment** | **Bank cards** | **Smart mobility (MIFARE) cards** | **Tags & Authentication** | **Readers** | **Mobile**

Post-Quantum Cryptography

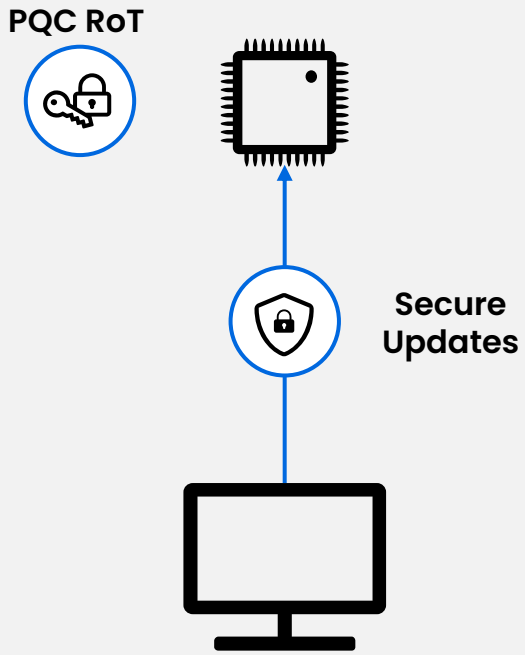


CURRENT	FUTURE
Public-Key Standards	**Post-Quantum Public-Key Standards**
RSA and ECC	Lattice-, Hash-, Code-based
Classical Computing	Classical Computing
Quantum-Safe	Quantum-Safe

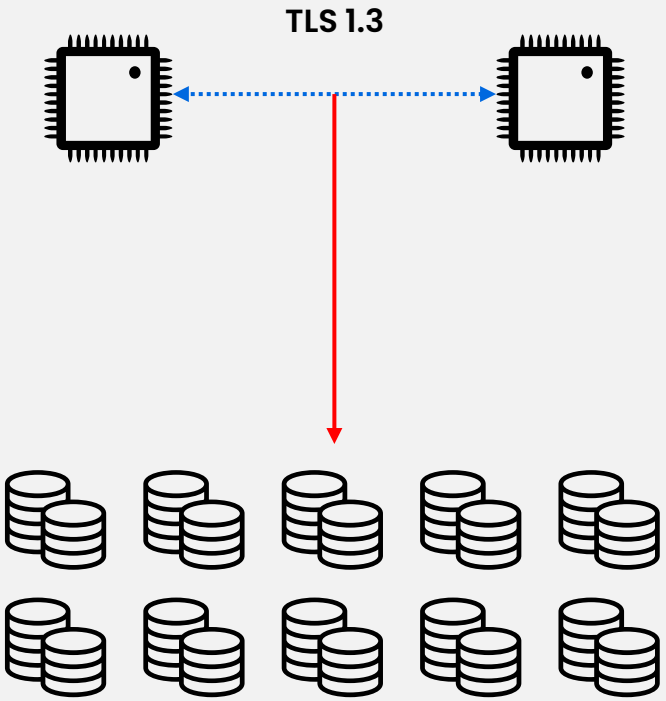
Why is Post-Quantum Cryptography relevant now?

Standards & Compliance

Devices in the field for decades



Store Now Decrypt Later



PQC standards

Key Exchange

Digital Signature

NIST
SP 800-208
Stateful Hash-Based Signature Schemes

2020

XMSS

LMS

2024

NIST PQC Standards

ML-KEM FIPS 203 (Kyber)

ML-DSA FIPS 204 (Dilithium)

SLH-DSA FIPS 205 (SPHINCS+)

Further NIST PQC Standards
Signature On-Ramp

2025-

FN-DSA FIPS 206 (Falcon)

HQC

Kpqc

2025-

NTRU+ | SMAUG-T

AIMer | HAETAE

2025-2028

Asia
More PQC standards expected

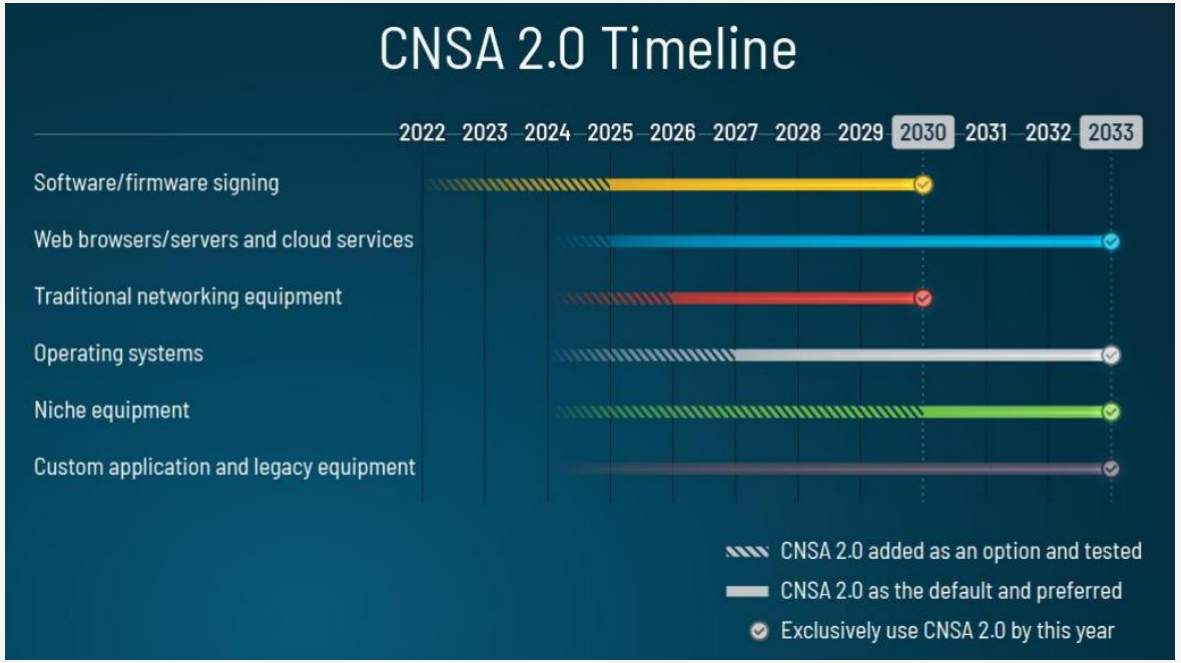
ISO
Push from EU to include additional schemes in ISO standard

2023-2030

FrodoKEM

Classic McEliece

PQC migration guidance



USA (NSA)

- NSA recommendation available
- Commercial National Security Algorithm Suite 2.0
- Begin transitioning immediately
- PQC FW signature supported by 2025
- PQC transition complete by 2030 using SW update

Germany (BSI)

- BSI first recommendation (English)
- BSI considerations (German)
- Expectation is that beginning of 2030s, a relevant quantum computer is available to be a threat for high-secure applications
- “QKD is only suitable for specific use cases”

France (ANSSI)

- PQC recommendations for security products
- “As soon as possible” when long-lasting protection is required
- Others to migrate to classic-PQC hybrid in 2025 – 2030
- Switch to PQC-only expected by 2030

The PQC migration path

Standards and regulations

Digital infrastructures

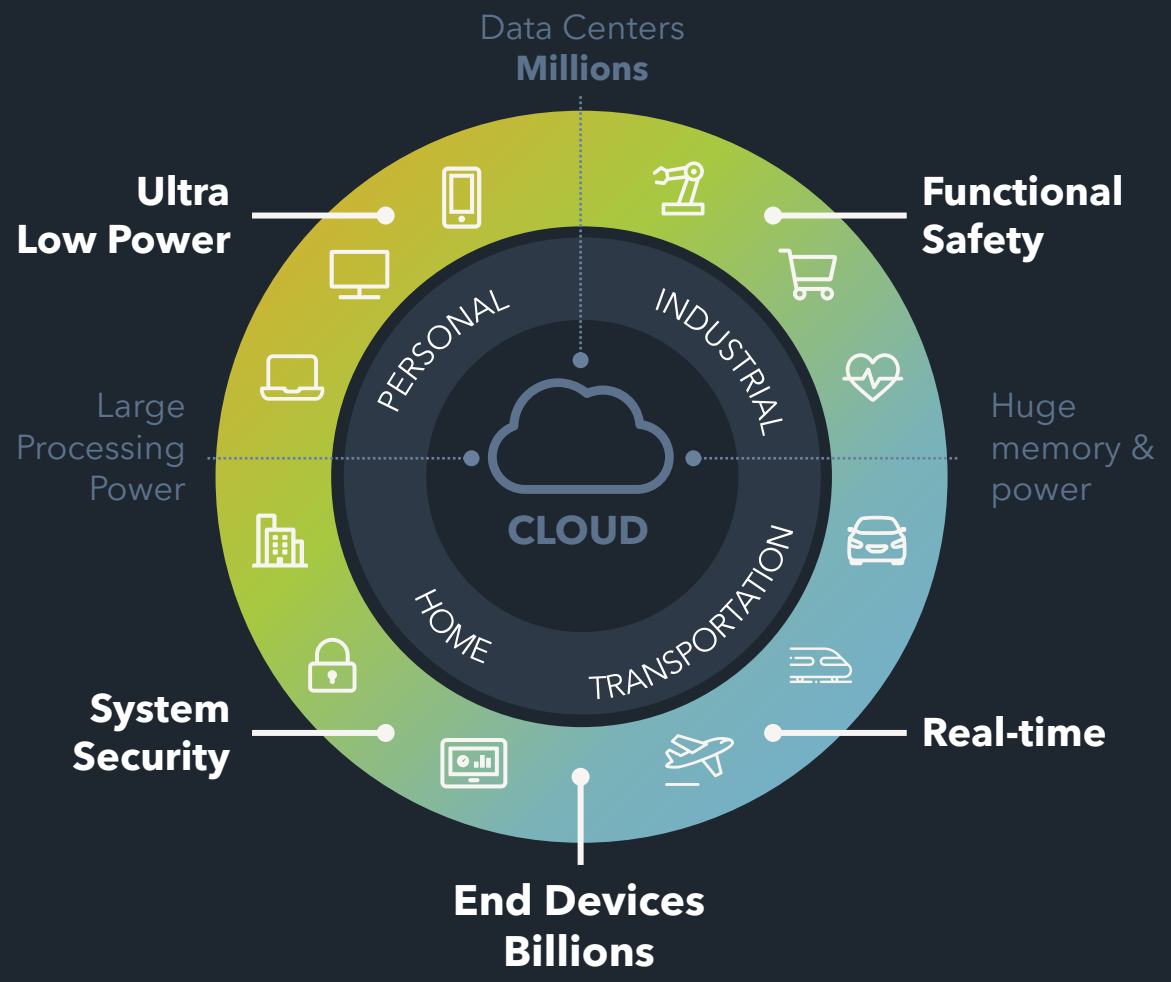
Protocols

Software

Hardware

Cryptography

Implementation Challenges for Embedded Devices



Data collection, processing and decisions at the edge
Devices securely connected to the cloud

No Silver Bullet
No one-size-fits-all solution. Some schemes are not suited for particular use cases & device constraints.

Cryptographic Keys
Orders of magnitude larger.

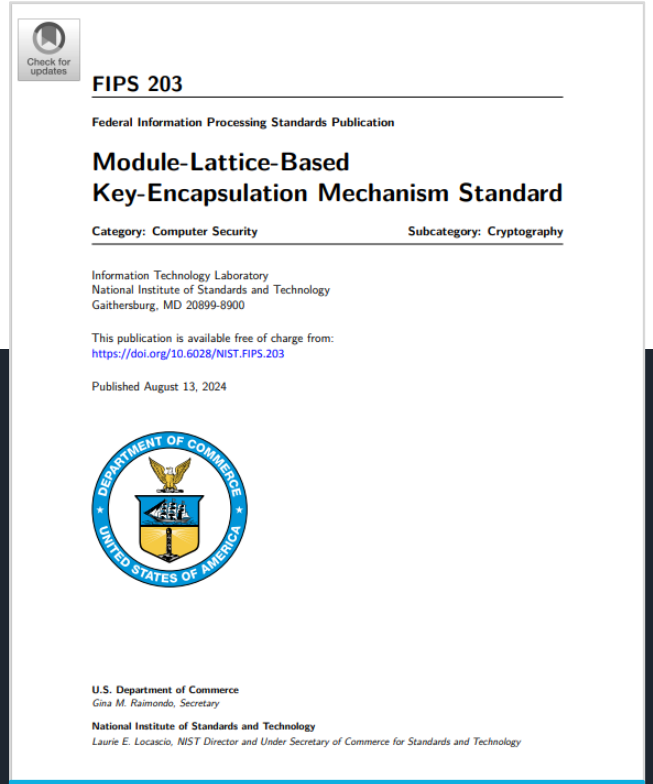
Bandwidth & Power
Larger signatures (up to 49.8KB).

Performance
Varies: some faster, some significantly slower.

Memory
Orders of magnitude more:
up to 100KB memory of RAM when executing.

Security
Implementation attacks: side-channel and fault injection.
Countermeasures can imply significant performance overheads.

New algorithms and standards



FIPS 203

Federal Information Processing Standards Publication

Module-Lattice-Based Key-Encapsulation Mechanism Standard

Category: Computer Security — Subcategory: Cryptography

Information Technology Laboratory
National Institute of Standards and Technology
Gaithersburg, MD 20899-8900

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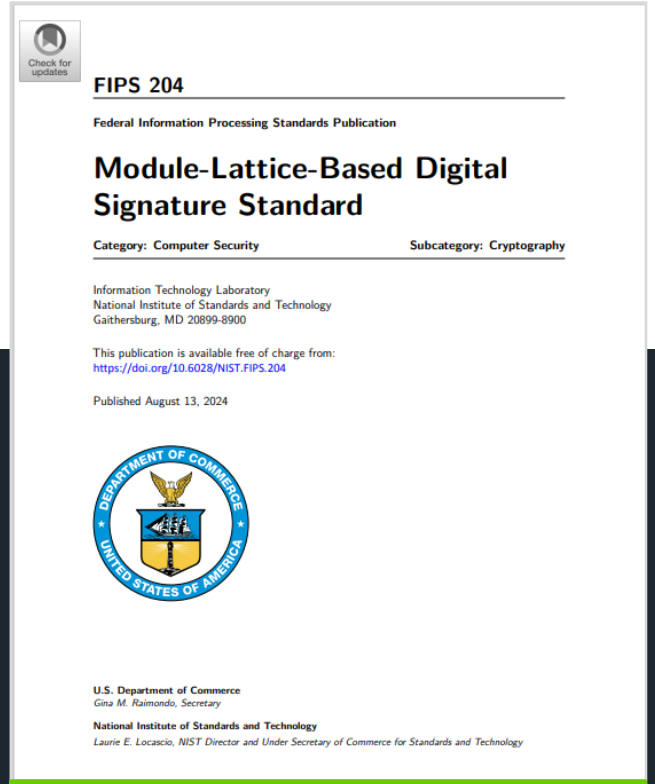
Key Exchange / Encapsulation



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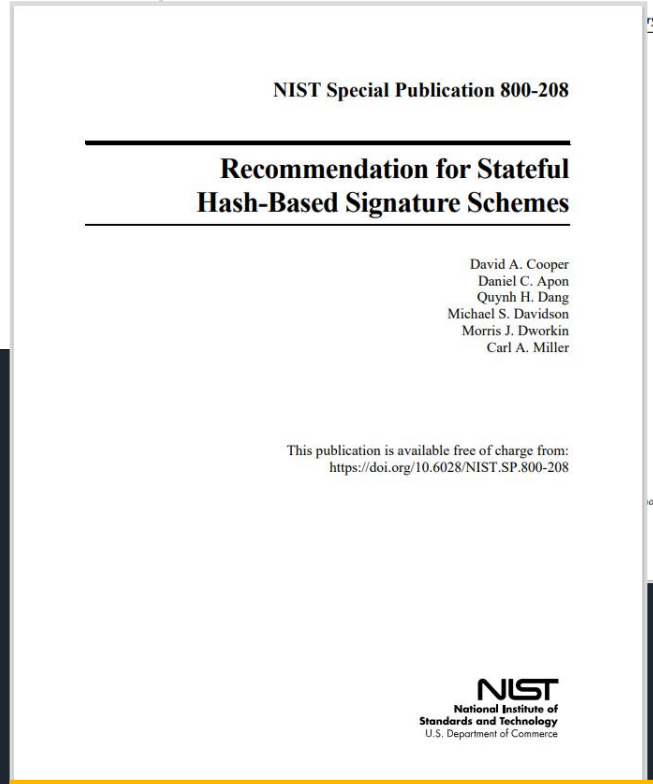

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Stateless Hash-Based Digital Signature Standard

NIST Special Publication 800-208

Recommendation for Stateful Hash-Based Signature Schemes

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Digital Signatures (software / firmware signing)

More ongoing and upcoming! FIPS 206, Round 4, On-Ramp, ISO, etc..

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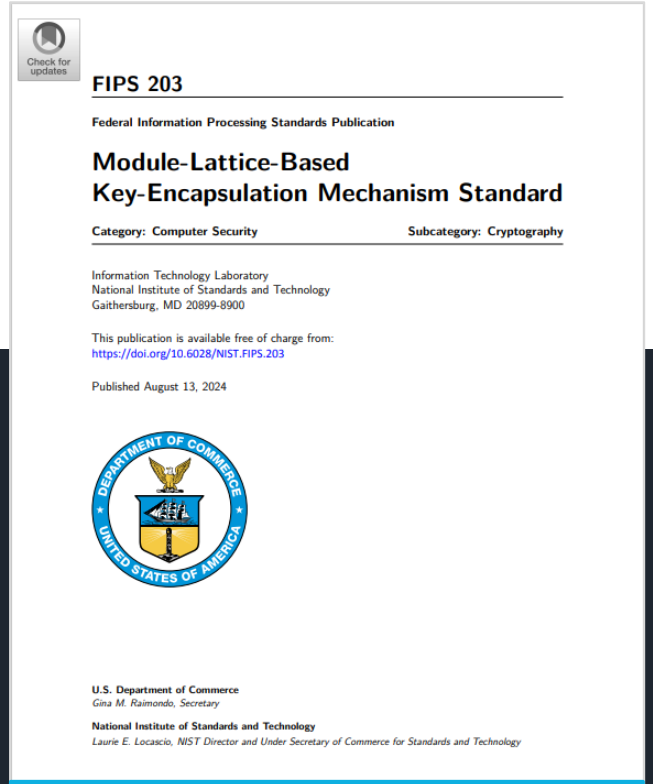
[3] SLH-DSA, https://nvlpubs.nist.gov/nistpubs/fips/nist.fips.205.pdf
[4] LMS / XMSS, https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-208.pdf

New algorithms and standards



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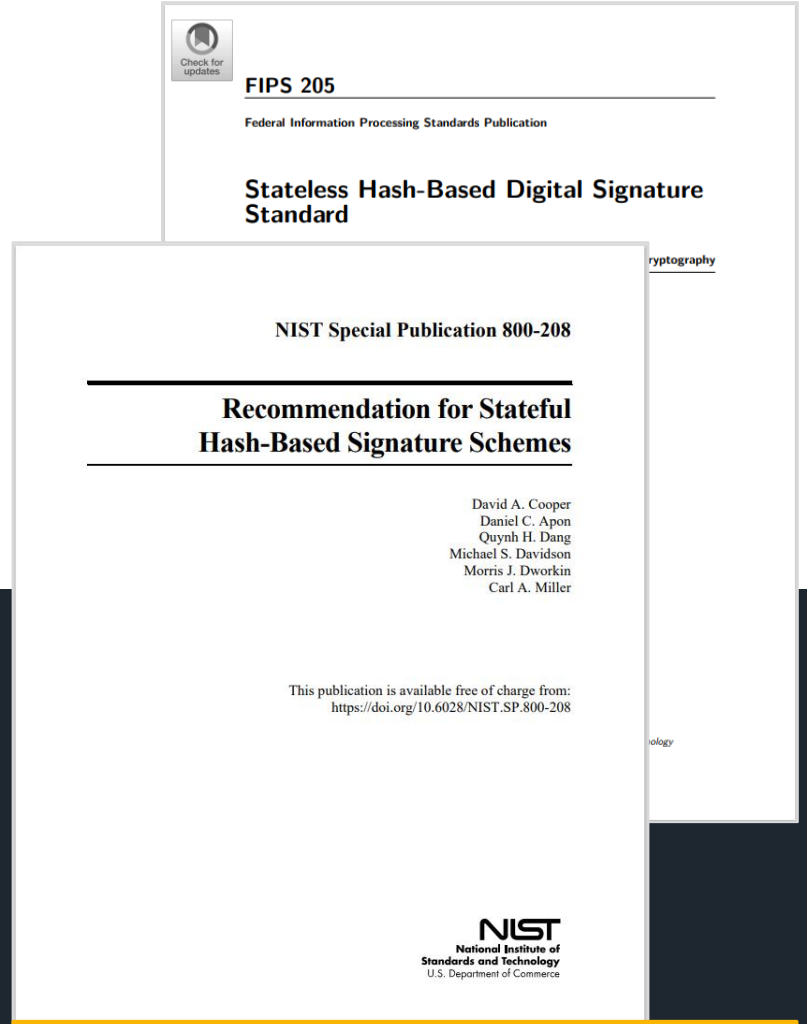

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[3] SLH-DSA, https://nvlpubs.nist.gov/nistpubs/fips/nist.fips.205.pdf
[4] LMS / XMSS, https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-208.pdf

From theory to practice: small-memory implementations

Can these implementations actually run on embedded systems?

		pqm4	
		Runtime	RAM
Dilithium-2	Sign	19 ms	50 kB
Dilithium-3	Sign	31 ms	69 kB
Dilithium-5	Sign	42 ms	123 kB

From theory to practice: small-memory implementations

Can these implementations actually run on embedded systems?

		pqm4		NXP PQC [A]		Slower	Smaller
		Runtime	RAM	Runtime	RAM	Runtime	RAM
Dilithium-2	Sign	19 ms	50 kB	61 ms	5 kB	3.2x	10.0x
Dilithium-3	Sign	31 ms	69 kB	119 ms	7 kB	3.8x	9.9x
Dilithium-5	Sign	42 ms	123 kB	168 ms	8 kB	4.0x	15.4x

All Dilithium parameter sets will fit on a device with ~8KB memory.

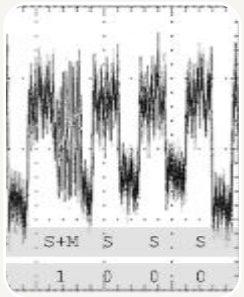
Price: factor 3 to 4 in performance
→ HW accelerators

[A] NXP PQC: Bos, J.W., Renes, J. and Sprenkels, A., 2022. Dilithium for memory constrained devices. In International Conference on Cryptology in Africa (pp. 217-235)

Resistance against physical attacks

Functional PQC is not sufficient to guarantee security

Countermeasures are necessary

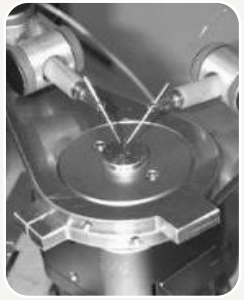


Side-channel attacks

- Power analysis (SPA, DPA)
- Electromagnetic analysis (SEMA, DEMA)
- Timing Analysis
- Photo-emission microscopy (high-end)
- Profiled, unprofiled and ML-assisted variants

Fault injection attacks

- Voltage or clock glitching
- Electromagnetic fault injection (EMFI)
- Body bias injection
- Laser fault injection
- Single and multi-shot scenarios



Invasive attack

- Focused Ion Beam (FIB) modifications
- Micro/Nano-probing of internal signals
- Signal forcing
- Delayering
- Reverse-engineering

Summary

Quantum security with PQC

- Not just “Harvest now, decrypt later”
- Devices in the field for decades
- Migration needed now!
- Official PQC standards are available

Embedded challenges of PQC

- Many layers/levels to the PQC migration
- Many use cases & applications
- Implementation challenges: memory, sizes, speed, physical security

Embedded security for PQC

- PQC alone doesn’t guarantee security
- Physical attack resistance for PQC implementations is essential for embedded and critical applications

Thank you!

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